Exp No. : 1
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BASIC NETWORKING COMMANDS

AIM :-

Basic Network Commands in Linux and Windows Operating Systems.

Basic Networking Command :-

arp - a :- ARP is short form of address resolution protocol.

host name :- This is the simplest of all TCP/IP Commands.

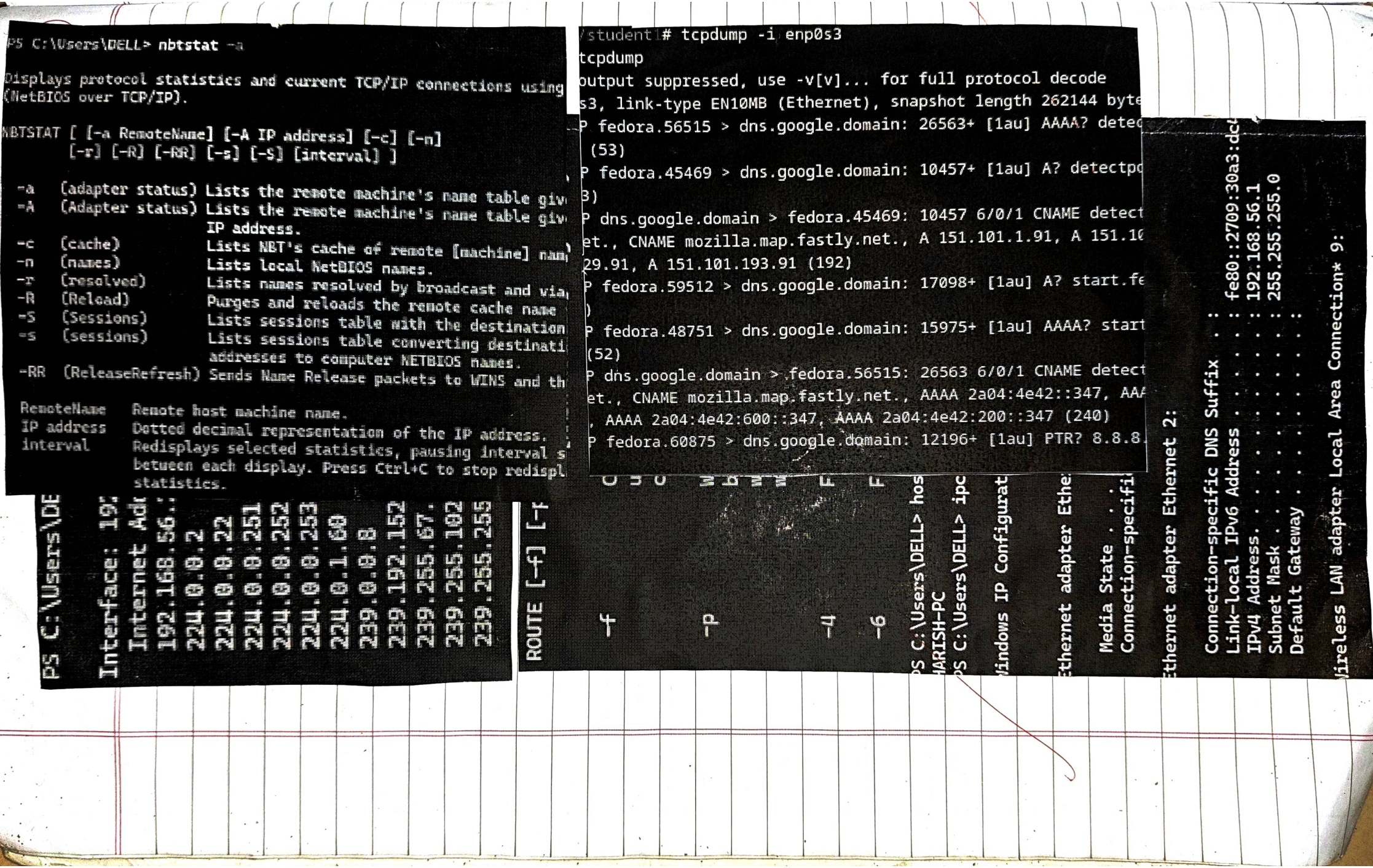
IP Config /all :- This command displays detailed Configuration information about connection including router, Gateway, DNS, DHCP.

nbtstat - a :- This command helps Solve problems with NetBIOS name resolution for NetBIOS over TCP/IP)

net stat - a :- This is a tool used to perform DNS look ups in DNS details.

ns look up :- This retrive the IP address of a domain name or DNS Server details.

Path Ping :- This is to trace the router to a destination and checks packet loss at each hop.



```
PS C:\Users\DELL> nbtstat -a

Displays protocol statistics and current TCP/IP connections using
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table giv
  -A   (Adapter status) Lists the remote machine's name table giv
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] nam
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via
  -R   (Reload)         Purges and reloads the remote cache name
  -S   (Sessions)       Lists sessions table with the destination
  -s   (sessions)       Lists sessions table converting destinati
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and th

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval s
               between each display. Press Ctrl+C to stop redispl
               statistics.
```

```
PS C:\Users\DE
Interface: 19
Internet Ad
192.168.56.
224.0.0.2
224.0.0.22
224.0.0.251
224.0.0.252
224.0.0.253
224.0.1.60
239.0.0.8
239.192.152
239.255.67.
239.255.102
239.255.255
```

```
ROUTE [-f] [-p
```

```
/student# tcpdump -i enp0s3
tcpdump
output suppressed, use -v[v]... for full protocol decode
s3, link-type EN10MB (Ethernet), snapshot length 262144 byte
P fedora.56515 > dns.google.domain: 26563+ [1au] AAAA? detec
 (53)
P fedora.45469 > dns.google.domain: 10457+ [1au] A? detectpo
3)
P dns.google.domain > fedora.45469: 10457 6/0/1 CNAME detect
et., CNAME mozilla.map.fastly.net., A 151.101.1.91, A 151.10
29.91, A 151.101.193.91 (192)
P fedora.59512 > dns.google.domain: 17098+ [1au] A? start.fe
)
P fedora.48751 > dns.google.domain: 15975+ [1au] AAAA? start
(52)
P dns.google.domain > fedora.56515: 26563 6/0/1 CNAME detect
et., CNAME mozilla.map.fastly.net., AAAA 2a04:4e42::347, AAA
, AAAA 2a04:4e42:600::347, AAAA 2a04:4e42:200::347 (240)
P fedora.60875 > dns.google.domain: 12196+ [1au] PTR? 8.8.8.
```

```
-f
-p
-4
-6
S C:\Users\DELL> hos
HARISH-PC
S C:\Users\DELL> ipc
Windows IP Configurat
Ethernet adapter Ethe
  Media State . . .
  Connection-specifi
Ethernet adapter Ethernet 2:
  Connection-specific DNS Suffix  . :
  Link-local IPv6 Address . . . . . : fe80::2709:30a3:dc
  IPv4 Address. . . . . : 192.168.56.1
  Subnet Mask . . . . . : 255.255.255.0
  Default Gateway . . . . . :
Wireless LAN adapter Local Area Connection* 9:
```

Ping :- This sends ICMP Packet to check whether a host is reachable and measures response time.

route :- Displays or changes the IP routing tables used by the System.

Linux Networking Commands :-

* IP :- This ip command is one of the basic command every administrator will need in daily work. This ip command can show address information manipulate rating plus display network various, integer and tunnels.

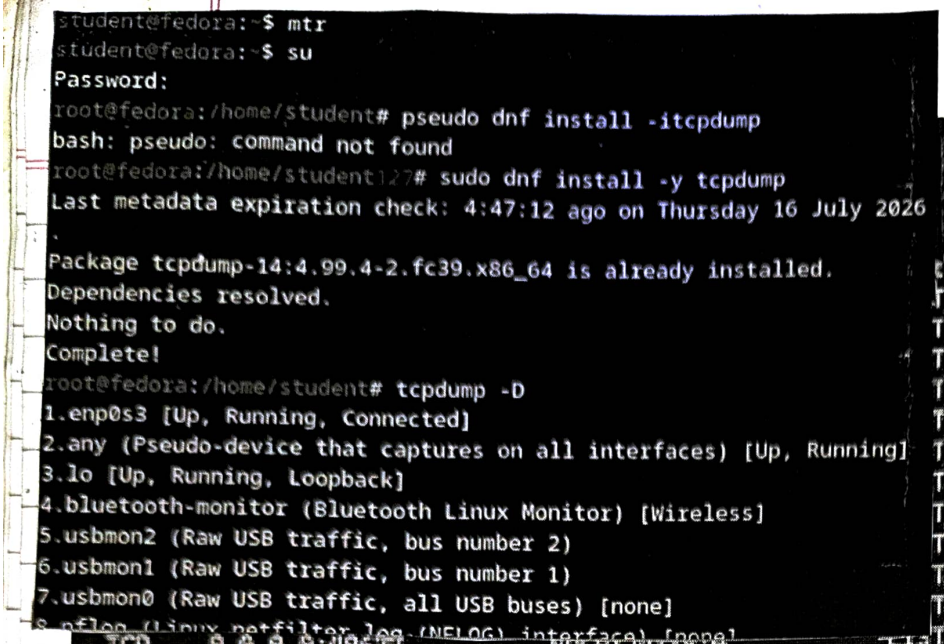
* if Config :- The if config command is a staple in many sysadmin's tool belt for configuration and trouble shooting.

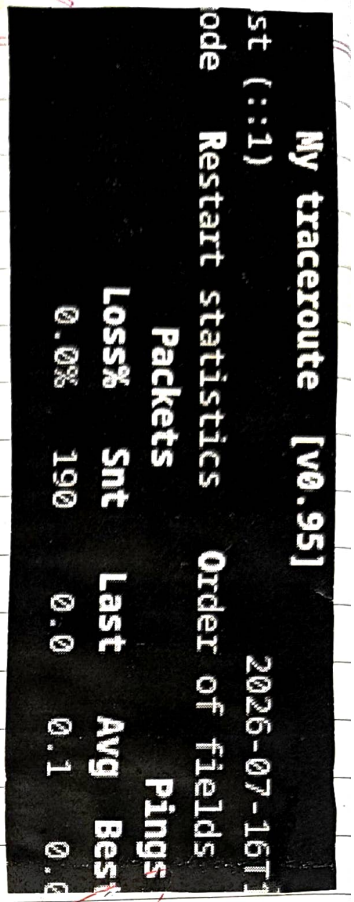
* mtr :- It is a program with a command-line interface the servers as a network diagnostic and troubleshooting tool.

Tcp dump :- The tcp dump command is designed for capturing and displaying packets.

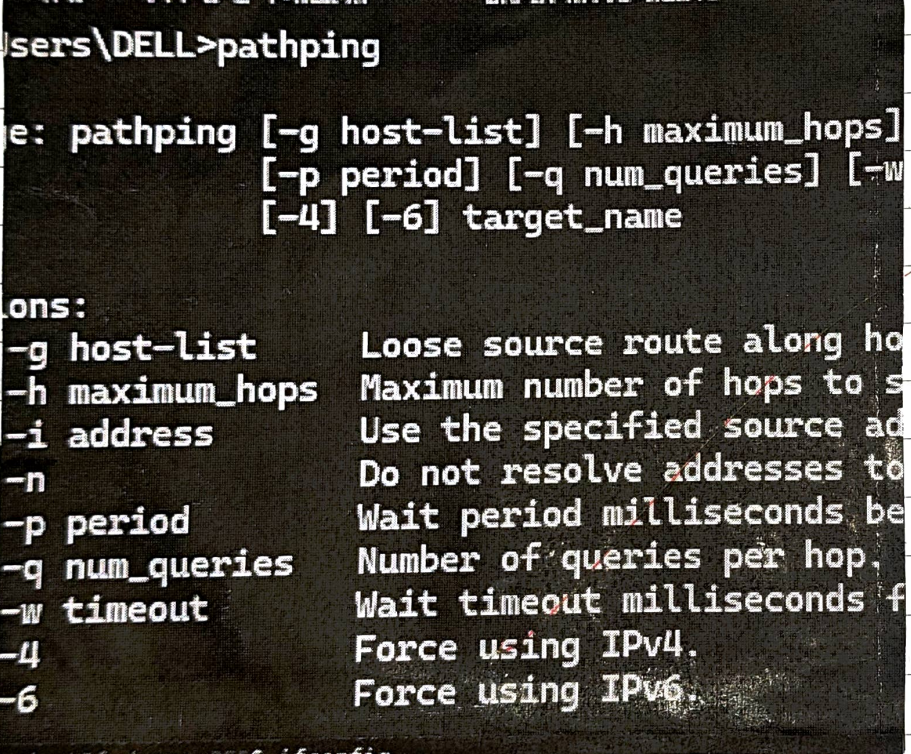
* Ping :- It is a tool that verifies IP-level connectivity to another TCP/IP Computer by sending Internet control message protocol echo request message.

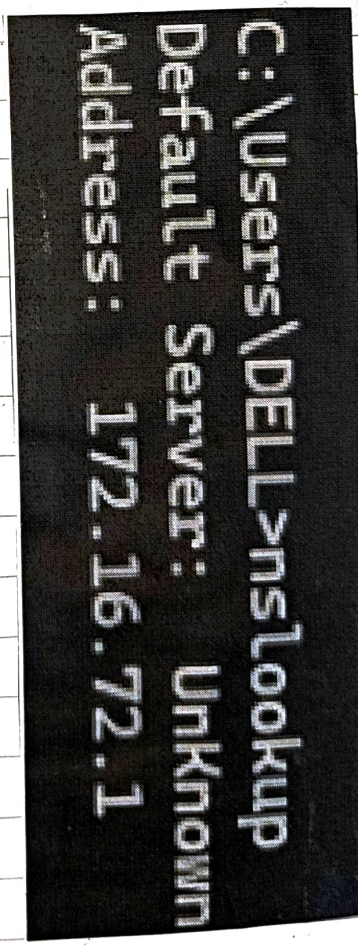
Result :- Basic networking commands in both Linux and windows operating system is successfully executed & verified.

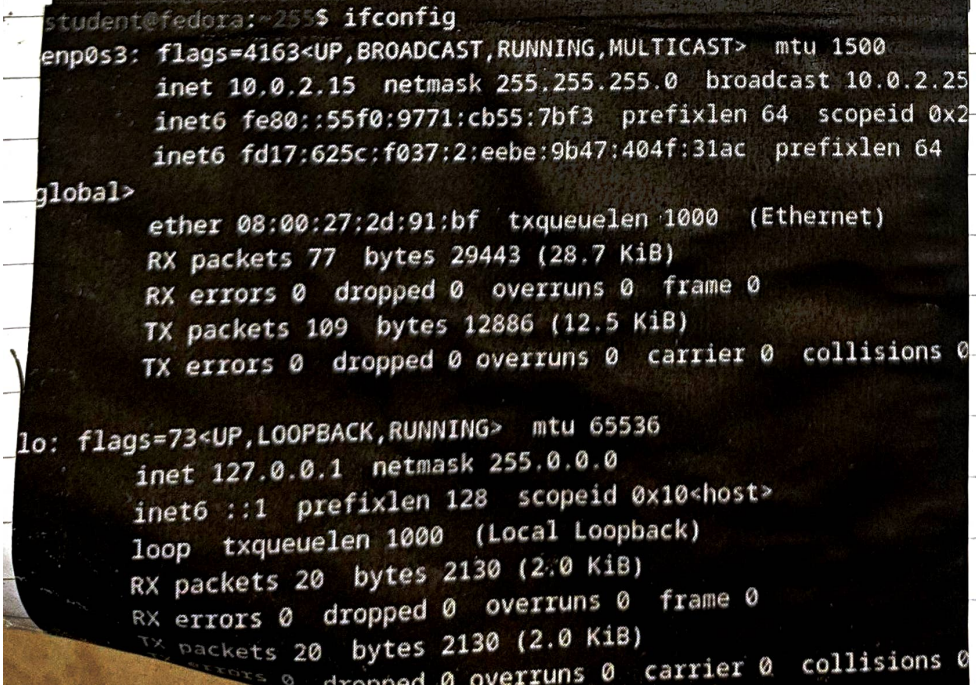

student@fedora:~$ mtr
student@fedora:~$ su
Password:
root@fedora:/home/student# pseudo dnf install -itcpdump
bash: pseudo: command not found
root@fedora:/home/student# sudo dnf install -y tcpdump
Last metadata expiration check: 4:47:12 ago on Thursday 16 July 2026
Package tcpdump-14:4.99.4-2.fc39.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
root@fedora:/home/student# tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses) [none]


My traceroute [v0.95]
Restart statistics Order of fields
Packets Pings
Loss% Snt Last Avg Best
0.0% 190 0.0 0.1 0.0


TCP 0.0.0.0:49677
TCP 127.0.0.1:27017 HDC0124118:0 LISTEN
TCP 127.0.0.1:49671 HDC0124118:49672 ESTABL
TCP 127.0.0.1:49672 HDC0124118:49671 ESTABL
TCP 127.0.0.1:49673 HDC0124118:49674 ESTABL


sers\DELL>pathping
e: pathping [-g host-list] [-h maximum_hops]
[-p period] [-q num_queries] [-w
[-4] [-6] target_name
ons:
-g host-list Loose source route along ho
-h maximum_hops Maximum number of hops to s
-i address Use the specified source ad
-n Do not resolve addresses to
-p period Wait period milliseconds be
-q num_queries Number of queries per hop.
-w timeout Wait timeout milliseconds f
-4 Force using IPv4.
-6 Force using IPv6.


C:\Users\DELL>nslookup
Default Server: UnKnown
Address: 172.16.72.1


student@fedora:~255$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.25
inet6 fe80::55f0:9771:cb55:7bf3 prefixlen 64 scopeid 0x2
inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac prefixlen 64
global>
ether 08:00:27:2d:91:bf txqueuelen 1000 (Ethernet)
RX packets 77 bytes 29443 (28.7 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 109 bytes 12886 (12.5 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 20 bytes 2130 (2.0 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 20 bytes 2130 (2.0 KiB)
dropped 0 overruns 0 carrier 0 collisions 0